

Installation & Setup

getting C++ running on Windows and Mac

Two things have to be on your machine before you can run C++: an **editor** to write the code in, and a **compiler** to turn that code into something the CPU can run.

IMPORTANT

VS Code is **only an editor**. It does not include a compiler. Installing VS Code alone will not run your C++ code — you have to install the compiler separately.

WHAT	WINDOWS	MACOS
Editor	VS Code	VS Code
Compiler	MinGW (g++), from the MinGW installer	Clang, from Xcode Command Line Tools
Command you will use	g++	g++



On Mac, `g++` is an alias for Apple's `clang++`. So the same commands work on both systems, and every command in this course will run on either machine.

Windows Setup

VS Code + the g++ compiler from MinGW

Step 1 — Install VS Code

- 1 Go to `code.visualstudio.com` and download the Windows installer.
- 2 Run the installer. On the setup screen, tick "**Add to PATH**" — it makes life much easier later.
- 3 Finish the installation and open VS Code once to confirm it launches.

Step 2 — Download MinGW

- 1 Search for **MinGW** and open the MinGW download page.
- 2 Click **Downloads** and download the installer file.
- 3 Run the installer and keep the default install folder, which is `C:\MinGW`.
Note this folder down — you need it in Step 4.
- 4 Click **Continue** and let it finish. The **MinGW Installation Manager** window opens automatically.

Step 3 — Select the g++ Compiler

The manager shows a list of packages. You only need the C++ one.

- 1 In the left panel, click **Basic Setup**.
- 2 In the list on the right, find `mingw32-gcc-g++` — this is the **g++ compiler**.
- 3 Right-click it and choose **Mark for Installation**. A tick appears next to it.
- 4 From the top menu, click **Installation** → **Apply Changes**.

- 5 Click **Apply** in the box that appears.
- 6 Wait for the download to finish, then click **Close**.



Mark only `mingw32-gcc-g++`. That single package gives you everything you need to compile C++ for this course.

Step 4 — Set the Environment Path

The compiler is installed, but Windows does not know where it lives yet. Remember PATH from the Files & Terminal notes — it is the list of folders Windows searches when you type a command.

- 1 Open **File Explorer** and go to `C:\MinGW\bin`.
- 2 Click the address bar at the top and **copy the path**.
- 3 Press the Windows key, type **environment variables**, and open **Edit the system environment variables**.
- 4 Click the **Environment Variables** button.
- 5 Under **System variables**, select **Path** and click **Edit**.
- 6 Click **New** and **paste the path** you copied.
- 7 Click **OK** on all three windows to close them.
- 8 **Close every open terminal and VS Code window.** PATH changes only apply to programs opened afterwards.

THE PATH YOU ARE ADDING

`C:\MinGW\bin`

Step 5 — Verify It Works

Open a **new** Command Prompt or PowerShell window and run:

```
COMMAND PROMPT
```

```
g++ --version
```

You should see a version number printed. If you do, the compiler is installed correctly and on your PATH.

macOS Setup

VS Code + Clang from Xcode Command Line Tools

Step 1 — Install VS Code

- 1 Go to code.visualstudio.com and download the Mac version.
- 2 Unzip it and drag **Visual Studio Code** into your **Applications** folder.
- 3 Open it once to confirm it launches.

Step 2 — Check If You Already Have a Compiler

Mac often ships with Clang already installed. Open **Terminal** (Cmd + Space, type "Terminal") and run:

```
TERMINAL
```

```
clang++ --version
```

If a version number appears, you already have a compiler — skip to Step 4.

Step 3 — Install the Command Line Tools

If the last command said "command not found", run this instead:

TERMINAL

```
xcode-select --install
```

- A dialog box will pop up. Click **Install** and accept the licence.
- Wait for it to finish — it is a large download.
- You do **not** need the full Xcode app. The Command Line Tools alone are enough, and far smaller.

Step 4 — Verify It Works

TERMINAL

```
g++ --version
```

You should see something beginning with **Apple clang version**. That is expected — as mentioned earlier, **g++** on Mac points to Clang.

VS Code Extension

the same on both systems

- 1 Open VS Code.

- 2 Open the Extensions panel — **Ctrl + Shift + X** on Windows, **Cmd + Shift + X** on Mac.
- 3 Search for **C/C++** and install the one published by **Microsoft**.
- 4 Restart VS Code so the extension loads cleanly.



This extension gives you IntelliSense — autocomplete, error squiggles and hover documentation. It does not compile anything on its own; the compiler you installed above still does that work.

Your First C++ Program

write it, compile it, run it

- 1 Make a folder anywhere for your code — for example `cpp-practice`.
- 2 In VS Code, go to **File** → **Open Folder** and select it.
- 3 Create a new file called `hello.cpp` and type the program below.

HELLO.CPP

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World";
    return 0;
}
```

Now open the terminal inside VS Code with **Ctrl + `** (Cmd + ` on Mac) and compile it.

STEP	WINDOWS	MACOS
Compile	<code>g++ hello.cpp -o hello</code>	<code>g++ hello.cpp -o hello</code>
Run	<code>.\hello.exe</code>	<code>./hello</code>

If **Hello World** appears in the terminal, your setup is complete.



Reading the compile command: `g++` is the compiler, `hello.cpp` is your source file, and `-o hello` tells it to name the output file `hello`. Leave off `-o` and you get a default name instead.

Shortcut — The C++ Snippet

stop typing the same boilerplate every time

You will write `#include <iostream>` and `int main()` hundreds of times in this course. A snippet lets you type three letters instead.

Setting It Up

- 1 Open the Command Palette — **Ctrl + Shift + P** on Windows, **Cmd + Shift + P** on Mac.
- 2 Type **Configure User Snippets** and select it.
- 3 In the list that appears, type **c++** and select **c++.json**.
- 4 Delete whatever is in the file and paste the code below.

- 5 Save with **Ctrl + S** (Cmd + S on Mac).

CPP.JSON

```
{
  "Print to console": {
    "prefix": "cpp",
    "body": [
      "#include <iostream>",
      "using namespace std;",
      "",
      "int main() {",
      "\t$",
      "\treturn 0;",
      "}"
    ],
    "description": "Basic C++ template"
  }
}
```

Using It

- 1 Open any **.cpp** file.
- 2 Type **cpp** and press **Tab** (or Enter when the suggestion appears).
- 3 The whole boilerplate drops in, with your cursor already inside **main()**, ready to type.



Two things worth knowing about that file. The outer `{ }` braces must be there — the snippet definition sits inside them, and VS Code will ignore the file if they are missing. And `$0` marks where your cursor lands after the snippet expands, which is why it sits inside `main()`.